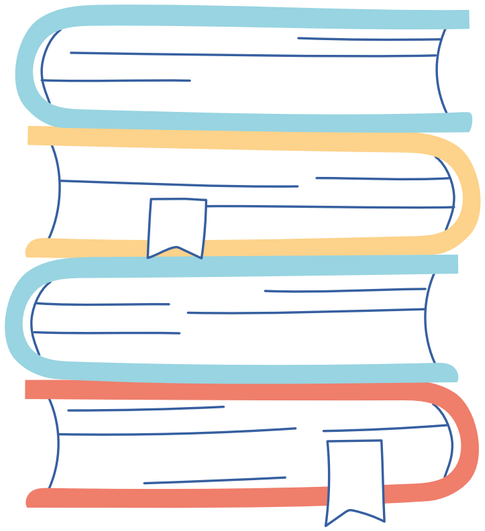
RECOMMENDATION DE LIVRE

Atelier II, IA

Charlotte Chanudet, Mahaut Galice, Marie Howet

SOMMAIRE



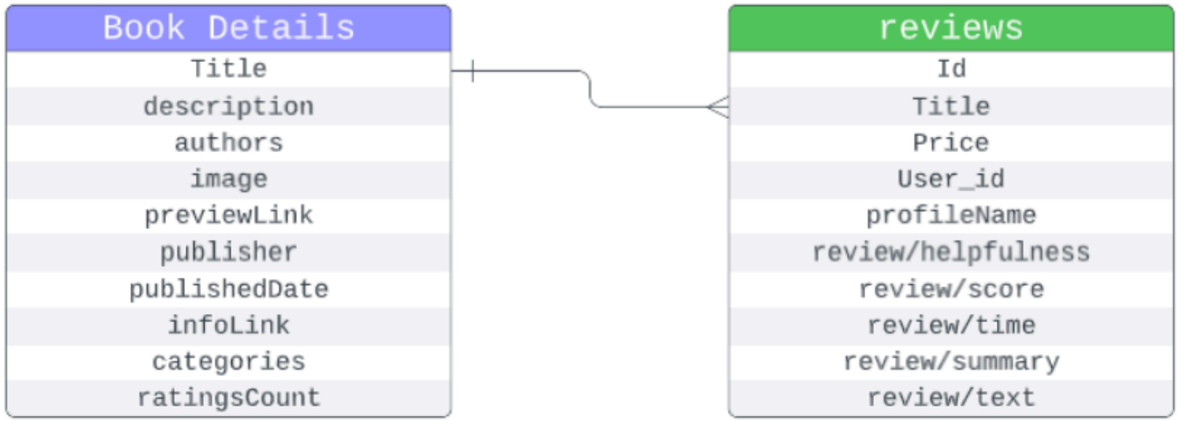
1. Explication des donnees
2. Modele par similarité
3. Modele GNN
4. Modele hybride
5. Analyse de nos résultats
6. Conclusion

1- LES DONNEES

- dataset kaggle : https://www.kaggle.com/datasets/mohamedbakhet/amazon-books-reviews
- 2 CSVs : Un pour les avis le second pour les données bruts des livres

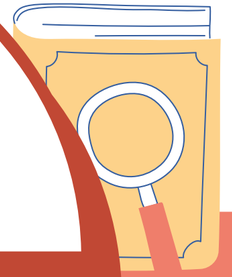
Traitement :

- Suppression des champs vide (144 000 lignes après tri)
- Filtrages des descriptions trop courte
- Filtrage des caractères ascii
- Filtrage des nombre minimaux d'interaction utilisateur-livre
- Filtrage sur la note minimale considérée comme positive.
- Encodage des identifiants



2- RECHERCHE PAR SIMILARITE

1. Objectif : recommander des livres sur base de similarité avec les descriptions d'autres livres
2. Encodage avec SBERT des resumes
3. Recherche KNN
4. Evaluation du modele et test unitaire pour visualisation
5. Avantages : Assez rapide, permet une premiere visualisation coherente
6. Probleme : Performance moyenne



2- RECHERCHE PAR SIMILARITE

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LIVRE REQUETE (index 2985)
   Titre       : CSS Mastery: Advanced Web Standards Solutions
   Auteur      : ['Cameron Moll', 'Andy Budd', 'Simon Collison']
   Categories  : ['Computers']
   Description : One of very few CSS books that focuses on teaching advanced-level CSS techniques CSS is a huge market—every web designer needs to know CSS to succeed in
 the modern web design marketplace Part of the "Solutions" series, and a follow-up from the best-selling Dan Cederholm book, Web Standards Solutions...

TOP 5 LIVRES SIMILAIRES
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  #1  Spring Into HTML and CSS
       Auteur      : ['Molly E. Holzschlag']
       Categories  : ['Computers']
       Similarite  : 65.8%
       Description : The fastest route to true HTML/CSS mastery! Need to build a web site? Or update one? Or just create some effective new web content? Maybe you just n
eed to update your skills, do the job better. Welcom...

  #2  HTML: The Complete Reference
       Auteur      : ['Thomas Powell']
       Categories  : None
       Similarite  : 61.7%
       Description : The Definitive Guide to HTML and CSS--Fully Updated Written by a Web development expert, the fifth edition of this trusted resource has been thoroug
hly revised and reorganized to address HTML5, the re...

  #3  250 HTML and Web Design Secrets
       Auteur      : ['Molly E. Holzschlag']
       Categories  : ['Computers']
       Similarite  : 55.2%
       Description : This value-priced guide by one of the Top 25 Most Influential Women on the Web delivers 250 solutions, workarounds, tips, and annoyance-busters that
 Web designers won't find anywhere else Offers 500 p...

  #4  HTML: A Beginner's Guide
       Auteur      : ['Wendy Willard']
       Categories  : ['Computers']
       Similarite  : 54.7%
       Description : Essential HTML skills—made easy! Thoroughly updated and revised, HTML: A Beginners Guide, Fifth Edition shows you, step by step, how to create dynam
ic websites with HTML. The book covers new HTML5 fea...

  #5  HTML for the World Wide Web, Second Edition (Visual QuickStart Guide)
       Auteur      : ['Elizabeth Castro']
       Categories  : ['Computers']
       Similarite  : 53.1%
       Description : Demonstrates the basics of HTML while explaining how to design Web sites, format text, add multimedia effects, and create forms, tables, lists, a
style sheets...
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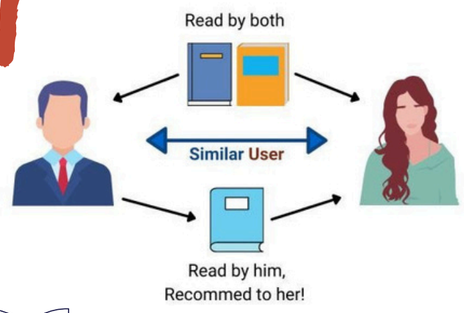
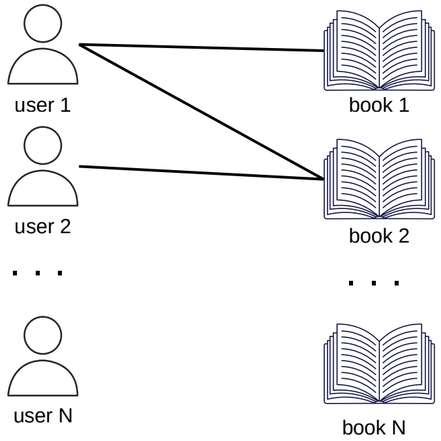
3- MODELE GNN : LIGHTGCN



1. Objectif : Recommander des livres sur base des interactions entre utilisateurs et livres.

2. Graphe biparti Utilisateur-Livre



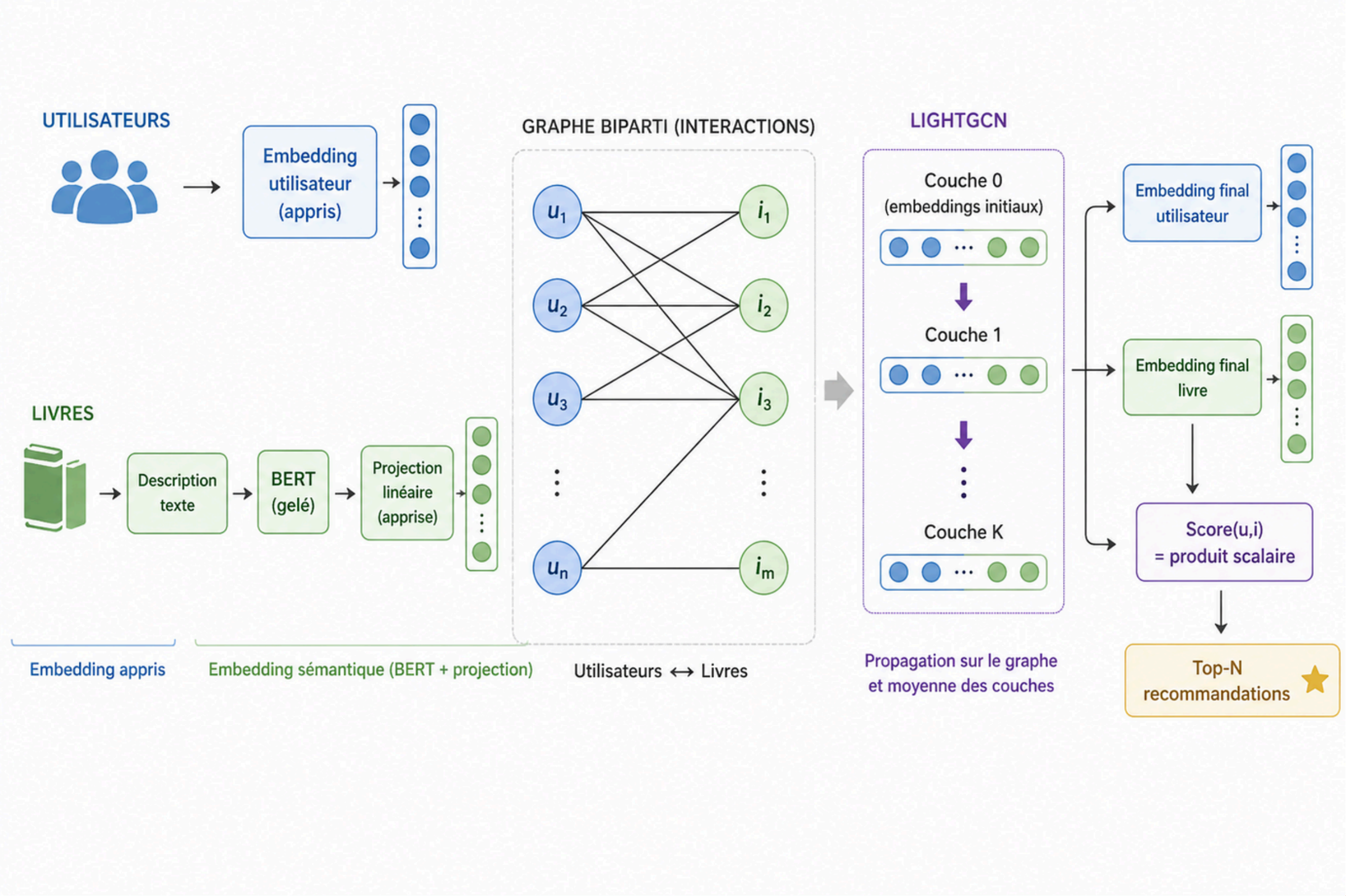
3. Principe de LightGCN $$e_u^{(k+1)} = \sum_{i \in N_u} \frac{1}{\sqrt{|N_u|}\sqrt{|N_i|}} e_i^{(k)}$$

4. Perte Bayesian Personalized Ranking

5. Avantages : Plus léger qu'un GCN et meilleur pour les systèmes de recommandations

6. Problèmes : Performance décevante + biais de popularité

4- MODÈLE HYBRIDE : LIGHTGCN + BERT



4- MODÈLE HYBRIDE : LIGHTGCN + BERT

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Recommandations
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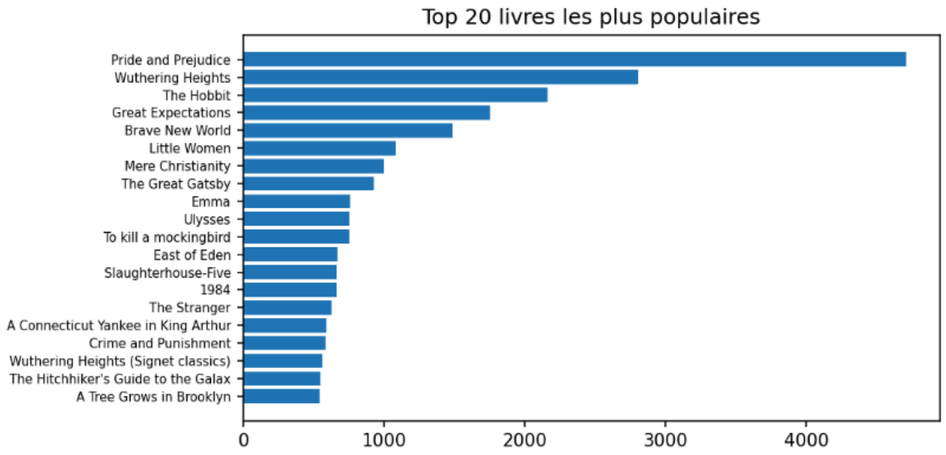
Utilisateur : A1N5FSCYN4796F

--- Livres lus (TRAIN) ---
Livres lus (train) — User A1N5FSCYN4796F : 4 livres
                                                title                  authors            categories
0                                Out of the silent planet (Avon)  ['Clive Staples Lewis']  ['College teachers']
1                                           A Death in Vienna         ['Daniel Silva']           ['Fiction']
2                                        Perelandra, a novel           ['C.S. Lewis']           ['Fiction']
3  OUT OF THE SILENT PLANET - The Perelandra Trilogy Book (1) One           ['C.S. Lewis']           ['Fiction']

--- Livres lus (TEST — vérité terrain) ---
Livres lus (test) — User A1N5FSCYN4796F : 2 livres
                                              title                 authors                categories
0                                Perelandra: A Novel           ['C.S. Lewis']               ['Fiction']
1  The Five Dysfunctions of a Team: A Leadership Fable  ['Patrick M. Lencioni']  ['Business & Economics']

--- Recommandations HybridLightGCN ---
User A1N5FSCYN4796F — Livres dans le test : 2 | Hits : 1/10 | Precision@10 : 0.10 | Recall@10 : 0.50
                                  titre                auteurs            catégories  score    hit
rang
1     The Hitchhiker's Guide to the Galaxy       ['Douglas Adams']           ['Science']  0.933  False
2               Stranger in a Strange Land  ['Robert A. Heinlein']           ['Fiction']  0.872  False
3                    The War of the Worlds         ['H. G. Wells']           ['Fiction']  0.869  False
4                      Perelandra: A Novel          ['C.S. Lewis']           ['Fiction']  0.868   True
5                           Lord of Light       ['Roger Zelazny']           ['Fiction']  0.863  False
6                   The Martian Chronicles        ['Ray Bradbury']           ['Fiction']  0.861  False
7                    That Hideous Strength          ['C.S. Lewis']           ['Fiction']  0.860  False
8                               The Hobbit    ['J. R. R. Tolkien']  ['Juvenile Fiction']  0.849  False
9                 The Left Hand of Darkness   ['Ursula K. Le Guin']           ['Fiction']  0.846  False
10                      The Battle for God     ['Karen Armstrong']          ['Religion']  0.846  False
```



5- RESULTATS

```
User A1N5FSCYN4796F — Test : 2 livres | K = 10
=========================================================
              Recall@10  Precision@10  Hits
Méthode
Aléatoire           0.0           0.0  0/10
BERT+KNN            0.5           0.1  1/10
GNN hybride         0.5           0.1  1/10
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--- Recommandations par méthode ---

[ Aléatoire ]
                                          titre  score hit
rang
1                                        Demian,  1.000   ❌
2                                   Finnegans Wake  0.999   ❌
3                                 The Four Feathers  0.999   ❌
4                                  The God Project  0.998   ❌
5                              Breakfast of Champions  0.998   ❌
6          The Mayor of Casterbridge (Signet classics)  0.997   ❌
7                                     The Sea Wolf  0.997   ❌
8     Interview With The Vampire SIGNED First Edition  0.997   ❌
9                         A Journal of the Plague Year  0.996   ❌
10                                        King Rat  0.996   ❌
```

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[ BERT+KNN ]
                                                                    titre  score hit
rang
1                                                          Perelandra: A Novel  1.000   ✅
2          Touching the Void: The True Story of One Man's Miraculous Survival  0.630   ❌
3                                                             Enduring Love  0.626   ❌
4                                                 Typee: Peep at Polynesian Life  0.625   ❌
5                                                Typee: A Peep at Polynesian Life  0.625   ❌
6                                                     Carry Me Across the Water  0.621   ❌
7                                                                  NEW EARTH  0.619   ❌
8                                                            The immoralist  0.615   ❌
9                                                            The Immoralist  0.615   ❌
10    Stranger in a Strange Land : ( Science Fiction Book Club Ltd Edition )  0.614   ❌
```

```
[ GNN hybride ]
                                        titre  score hit
rang
1      The Hitchhiker's Guide to the Galaxy  0.933   ❌
2                 Stranger in a Strange Land  0.872   ❌
3                      The War of the Worlds  0.869   ❌
4                        Perelandra: A Novel  0.868   ✅
5                              Lord of Light  0.863   ❌
6                      The Martian Chronicles  0.861   ❌
7                       That Hideous Strength  0.860   ❌
8                                  The Hobbit  0.849   ❌
9                   The Left Hand of Darkness  0.846   ❌
10                         The Battle for God  0.846   ❌
```

GNN hybride est le meilleur compromis

6- CONCLUSION

- Objectif atteint, resultats logiques avec des livres recommandes coherents avec les livres lus
- Dataset potentiellement a changer
- Approche hybride quand meme bien meilleure
- Voir si un autre encodeur aurait pu donner de meilleurs resultats